Document Control — git as the document control system for RDM’s record

Document ID	DC-001
Revision	5
Date	2026-11-11
Status	Controlled in git

Contents

1 Purpose 3
2 The system 3
3 Controls 3
4 Verification 4

Id: `DC-001` · Revision: `5`

1 Purpose

States how the controlled records in RDM's repository are controlled: RDM's own — this Design History File, its computer software validation (CSV-001), its spec (`.pi/spec/`), its tests and the source that meets them — and the worked example's, the computer software validation of Löyly's pi harness (`examples/loyly-pi-harness/`, CSV-HAR-001). This is the one Part 11 document for both: they share this repository, its history and its review. RDM keeps its record **in git, with GitHub as the service provider**; this statement maps each applicable 21 CFR Part 11 control to the mechanism that meets it, and it must pass gap analysis against RDM's own shipped `part11_document_control` checklist (item-05-004 — the same checklist any downstream project audits with is the one this record is held to).

2 The system

The master copy of every controlled record is the repository's default branch. Drafts are branches; review and approval happen on the pull request; releases are tags; the revision history is the git history itself. Planning artifacts (tasks, issues) are coordination, never part of the record (`docs/about/plan-vs-record.md`).

3 Controls

- **Validation** [[P11:11.10a]] — RDM and this record system are validated as CSV-001 records (computer software assurance, on a documented risk assessment). The controls are enforced configuration and code, exercised on every change: the CI pipeline (`design-controls.yml`) and the branch ruleset run RDM's own release gate and graph validation, verified by RDM's own acceptance tests, with every change independently reviewed on its pull request. Altered records are discernible by construction — any change produces a new SHA.
- **Copies** [[P11:11.10b]] — accurate and complete copies in human-readable and electronic form: the rendered documentation site and the rendered traceability matrix are generated from the record by `rdm render`; a `git archive` at any tag reproduces the complete electronic record set.
- **Retention and retrieval** [[P11:11.10c]] — every clone is a full replica of the history; release tags identify record baselines; the GitHub-hosted repository is the retained store. Retention period: the full history is retained for as long as RDM is maintained, and release baselines (tags and their evidence bundles) for no less than five years after the release they record. An organization-controlled mirror is an open action (audit NC-6).

- **Access** [[P11:11.10d]] — write access is limited to authorized individuals through GitHub organization membership with two-factor authentication; the audit log records permission changes.
- **Device checks** [[P11:11.10h]] — every writing client is authenticated to an account-bound credential (SSH key or token issued to the individual's 2FA-protected account); anonymous or unauthenticated input cannot reach the record.
- **Signature accountability** [[P11:11.10j]] — the contributor policy (`AGENTS.md` and this statement, both controlled documents) states that a pull-request approval is the approver's electronic signature and that the approver is accountable for what it approves; the approver is named on the pull request.
- **Audit trail** [[P11:11.10e]] — the git history: secure (SHA-chained), computer-generated, time-stamped, author-attributed on every action; a later change never obscures an earlier entry, and the trail lives exactly as long as the record it belongs to.
- **Sequencing** [[P11:11.10f]] — the enforced order of steps: CI blocks merge until the acceptance tests, verification, the release gate and graph validation pass, in that order; the reviewer of each pull request checks a requirement's scenario is in the spec before the code that meets it.
- **Authority** [[P11:11.10g]] — only authorized reviewers can approve a pull request into the default branch, and the approver must not be the change's author (enforced by the ruleset's required review). The branch controls are configuration code in this repository — `.github/rulesets/controlled-record.json` (pull-request review, verified commit signatures, required checks, no history rewriting) and `repo-settings.json` — applied and drift-checked with `scripts/apply-repo-controls.sh [--check]`.
- **Training** [[P11:11.10i]] — contributors (human or agent) are directed through `AGENTS.md` and `CLAUDE.md` before changing the record; agent sessions load them at session start.
- **Systems documentation control** [[P11:11.10k]] — this statement, the contributor guide and the CI workflow are themselves version-controlled in this repository and change only through the same approved path they describe.
- **Signature manifestation** [[P11:11.50]] — the pull-request review records the approver, the UTC timestamp, and the meaning of the signing (APPROVED — reviewed and approved for the record); the merged, reviewed PR is RDM's approval record by design (no duplicate sign-off blocks exist in the DHF).
- **Signature–record linking** [[P11:11.70]] — a review is bound to the exact commit SHA it approves; content-addressing makes a signature inseparable from what was signed.
- **Signature uniqueness** [[P11:11.100]] — one GitHub account per individual, two-factor authentication, accounts never reused or reassigned.

4 Verification

`rdm gap part11_document_control dhf/documents/document_control.md` must exit zero; the item-05-004 acceptance test executes exactly that, so this statement can never silently drop a control.